Non-invasive Optogenetics with Ultrasonic Waves and Sonoactive Nanoparticles

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***Abstract*— This study explores a new approach to optogenetics using ultrasonic waves and sonoactive nanoparticles to non-invasively control and monitor neural activity. Traditional optogenetic techniques are limited in their applicability and pose risks to subjects due to the necessity of invasive procedures. The main goal of this study is to propose a method that replaces these invasive techniques, allowing for precise control and monitoring of specific cells or tissues without invasive procedures. Sonoactive nanoparticles, activated by ultrasound, emit light to enable optogenetic regulation in the brain. This method not only improves targeting precision but also significantly reduces invasiveness, offering a valuable alternative for advancing the application of optogenetic technology in neurological research and therapy.**

Keywords*— *optogenetics, sonoactive nanoparticles, upconversion nanoparticles, blood-brain barrier

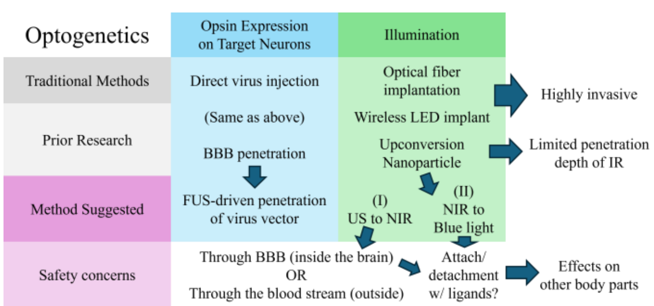

Fig. 1. Summary of Ideas. A brief summary of pipeline and comparison with traditional methods and prior research.

I. Introduction

A. Background

Optogenetics is an elegant approach for precisely controlling and monitoring the biological functions of a cell, a group of cells, tissues, or organs with high temporal and spatial resolution using optical systems and genetic engineering technologies. The field has evolved to precisely control neurons and decipher neural circuitry, achieving significant accomplishments in neuroscience. [1] This technology is applicable to optical systems already present in the clinical-diagnostic field and has shown excellent effects on biological investigations and therapeutic strategies. Recently, various biomedical applications of optogenetics have been explored, including in ophthalmology, bone repair, heart failure recovery, post-stroke recovery, tissue engineering, and regenerative medicine (TERM). [2]

B. Problem Statement

Optogenetics is a powerful tool for precisely controlling neural activity through light stimulation. However, traditional optogenetic methods require invasive procedures such as direct brain injections of viral vectors and fiber/electrode implantations, posing significant risks and limitations in research and clinical applications. Direct injections and fiber/electrode implantations are invasive and may lead to significant tissue damage and morbidity. The neural networks being studied are thus subject to disruption and potentially detrimental functional changes. [3] These issues limit the broader application of optogenetics in long-term studies and therapeutic interventions in the human brain. Therefore, a non-invasive alternative that provides the same level of precision without these risks is necessary.

C. Objectives

This study utilizes ultrasonic waves and sonoactive nanoparticles instead of invasive procedures. The goal is to develop a non-invasive optogenetic method that precisely controls and monitors neural activity. The specific objectives are as follows:

- Replace fiber implantations and direct injections of viral vectors by using nanoparticles
- Employ Focused Ultrasound (FUS) to target these nanoparticles to specific brain regions.
- Validate the efficiency and precision of this method through experimental research.

D. Research Methodology

The proposed method utilizes FUS and two types of nanoparticles: Ultrasonic to Near-Infrared Nanoparticles (TD-NP) and Upconversion Nanoparticles (UCNP). TD-NPs emit near-infrared light in response to ultrasonic stimulation, which is then converted to blue light by UCNPs, essential for activating light-sensitive opsins. This approach allows for precise, non-invasive modulation of neural activities.

II. Prior Works

A. Opsin delivery through BBB

Non-invasive delivery of opsin proteins is achieved by penetrating the blood-brain barrier (BBB) with viral vectors from the bloodstream to the brain. Strategies to penetrate the BBB, which blocks external substances between the blood and the brain, are required. Various methods, including the use of osmotic agents and capsid engineering, have been researched to increase the BBB's permeability. [4] However, these techniques are invasive or fail to enhance permeability in targeted areas.

Recent studies have utilized focused ultrasound (FUS) to increase BBB permeability. [5-6] FUS allows viral vector and nanoparticle deliver into the brain. [7] Additionally, FUS can enhance permeability in targeted specific area.

B. Light Excitation

As an alternative to optical fibers, microLED-based RF wireless implant devices have been conducted. [8-9] Although wireless precise control is available, it is still highly invasive. Non-invasive optogenetic activation using infrared (IR) has been studied, due to deep penetration of IR within the optical window (650 – 1350 nm) [10] Methods such as red shift [11], two-photon excitation [12] and upconversion nanoparticle (UCNP) [13] have been studied.

UCNP absorbs IR lights and emits lights with shorter wavelengths. This study showed that UCNP can activate ChR2 opsin in vitro and in vivo, from IR light outside the skull. [13] However, methods using infrared light face challenges penetrating deeper than 5mm, due to scattering and intensity reduction caused by the skull. Furthermore, non-invasive delivery of UCNP is needed.

III. Methods

We now propose a novel framework for injecting AAV-mediated opsin DNA, terbium-doped nanoparticles (TD-NP) and UCNP into and stimulating the target brain area. We first propose the overall method and give the calculations for specific dose and nanoparticle treatment. Finally, we state rooms available for refinement and experimental verification. The overall methodology is shown in Fig. 2.

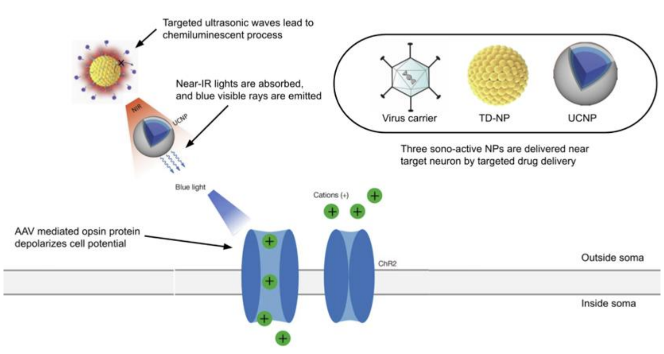

Fig. 2. Overview of Methodology. The overall method from mixture injection and ultrasonic stimulation of TD-NP to opsin activation with blue light is depicted in the image. Graphics are borrowed from [13-15].

A. Nanparticle Preparation and Characterization

1) TD-NP Nanoparticles

TD-NPs that emit NIR light upon ultrasonic stimulation are synthesized via a sol-gel method. These nanoparticles exhibit strong piezoelectric properties. The average size is maintained at 40-50 nm to balance tissue penetration and minimize immune response. [13] TD-NP is prepared at 100mg/ml, which is optimal for ensuring dispersion and minimizing aggregation in targeted brain regions. These concentration levels are similar to those used in Iodine-based contrast media used for CT scans or X-rays, and have the need to be proven for safety with medial concentrations, as stated afterwards in the experiments.

2) UCNP Nanoparticles

NaYF4/Tm-based upconversion nanoparticles (UCNPs) convert NIR light to blue light with high efficiency. For ChR2 activation, we are going to use blue-emitting NaYF4 nanocrystals co-doped with Yb^{3+}/Tm^{3+}, as given in the original study. [5]

UCNPs are synthesized at 90 nm to facilitate diffusion and interaction with TD-NPs and to efficiently break through BBB. Concentrations are set at 1mg/ml to match the TD-NP concentration ratio to 100:1 and ensure balanced interaction and upconversion efficiency. However, using different concentrations of TD-NPs and UCNPs may result in better overall performance, depending on factors such as the efficiency of NIR light emission by TD-NPs and the absorption and upconversion efficiency of UCNPs, and is left as a parameter that needs to be experimentally verified.

The UCNP nanoparticles have molecularly tailored structures, as given in prior study, that can act as ligands that combine to cellular proteins in the surface of the target soma. One example can be cocaine analogues(CFT, WIN 35) or propane derivatives(β-CFT) that are used as ligands that have high affinity with dopamine transporters(DATs).

B. Injection Methodology

1) MRI-Guided Focused Ultrasound (FUS)

We use MRI-guided focused ultrasound (FUS) to precisely target and open the blood-brain barrier (BBB) in specific brain regions such as dopamine pathways. The procedure is as follows: The patient is positioned in an MRI scanner. MRI is used to locate the target area, and FUS is applied to transiently open the BBB in the target area using an increased intensity of $3W/cm^2$ at 1MHz, balancing penetration depth and tissue safety. Below calculations show that less intensity is not enough to reach needed amount for effective stimulation.

Immediately after or simultaneously with the FUS application, the AAV vector is injected intravenously. The disrupted BBB allows the AAV to enter the targeted brain region. Inject TD-NP and UCNP nanoparticles intravenously. The AAV vector enters the neurons in the targeted brain region and delivers the opsin DNA. The neurons then express the opsin protein, making them sensitive to light stimulation.

2) Nanoparticle/AAV Injection

The nanoparticles are directed to the targeted brain regions through the temporarily opened BBB. We aim for 50 ml of brain tissue in the specific pathways rather than the entire brain volume. The nanoparticles are contained in a liposome that can be delivered to target areas, and crosses the BBB due to the invasion of the BBB. The nanoparticles, after delivery move towards the target neuron, for the molecules tailed in the surface of the nanoparticles are affine towards specific transporters in the surface of the target soma. By this way the nanoparticles can stay put near target somas, and directly stimulate surface potential. [16].

The BBB typically restores itself within a few hours after the FUS application, preventing further entry of substances from the bloodstream.

C. Stimulation Methodology

First the TD-NPs are stimulated with FUS, and with a piezoelectric effect that undergoes oxidation and results in a chemiluminescent process occurring. [14] The light that is emitted from this process has a wavelength near infra-red light, with the highest intensity near 650-700nm. The ultrasonic stimulation has a spatial resolution of few centimeters in the human brain [17], and therefore the TD-NP stimulation will be able to occur few centimeters near the target area.

This NIR wave is transmitted to nearby UCNP particles, where NIR is converted to blue light in real-time with an efficiency of 2.5% [5]. The highest conversion rate is when the NIR light has a wavelength of around 980nm. We can observe that the wavelength with the maximum emission rate is slightly different, around 250nm difference, and we believe that there is room for refinement to reduce the gap between the two. Because the NIR wave as a penetration rate of 2.2cm, and a spatial resolution of 1.46mm [13], we believe that the upconversion will be made under resolutions of 1mm near the soma (as molecular tails have chemical-specific targeting), within the boundary of 2cm of IR penetration.

Finally, the near-blue wavelength light stimulates the opsin proteins to create stimulation in the neurons. This leads to passive stimulation of experimental or clinically interested neuron pathways and gives same effects as invasive optogenetics.

The two-step stimulation is an important part in this methodology, for two-step stimulation is needed for low-to-high resolution targeting. FUS and TD-NP stimulation creates a NIR boundary of 2cm in diameter, such as Fig. 3. This low-resolution targeting enables potential UCNP positions to be stimulated with IR, but which of course will only be stimulated near each neuron, where UCNP has high affinity with. Then the high-resolution targeting takes effect, where only neurons with opsin proteins get stimulated from the near-blue light. The low-resolution step is crucial to assure that light is stimulated near path neurons, and the high-resolution step is crucial to assure that only neurons with opsins are targeted with this method.

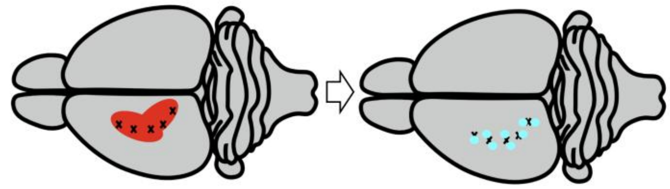

Fig. 3. Graphical Representation of *Low to High Resolution Targeting*. Regions to target are marked with "X" signs. Then left side represents US-induced NIR rays, which illuminate relatively broad area in comparison to NIR-induced blue rays on the right side. This process we named the *low to high resolution targeting*.

D. Calculations for Concentration and Intensity

We now show explicit calculations that set the appropriate intensity for ultrasonic waves and prove that with appropriate nanoparticle ratio, we can achieve the threshold for stimulation.

1) Required Brain Concentration

We target a specific region: assume 50ml of brain tissue for targeted pathways. Our desired concentration is 1mg/ml in the brain tissue for UCNP and 100mg/ml in the brain tissue for TD-NP. The total required dose in the targeted region would be 1mg/ml×50ml=50mg. The below calculations are done for UCNP nanoparticles only, for brevity. The TD-NP calculations are done similarly, but 100 times scaled up.

2) Injected Dose Calculation

We assume that 1% of the dose will reach the targeted area as a strong lower bound. This is because the injection is done intravenously, where the blood runs through the brain continuously and we will have to capture the drug dose with extremely small BBB openings.

3) Efficiency of Upconversion and Stimulation

As stated in [5], UCNP converts NIR to near-blue light with 2.5% ratio. [5] also states that the minimum threshold for stimulating neurons are 0.1mW/mm^2, and therefore we calculate the minimum threshold of NIR intensity to be 0.4W/cm^2.

The TD-NP process gives a downconversion intensity ratio (=intensity of NIR/intensity of FUS) of 2.5×10^{-3} scale, calculated from the experiments of prior study [13] where input is ultrasonic waves of $8\times10^{7}\ \text{W/cm}^2$ and outputs are counts of photons of 650nm wavelength. If this downconversion ratio applies to our concentration near the target neuron area, we calculate that there should be at least N times as many TD-NPs than UCNPs in the solution, where N is calculated as below:

$$N = \frac{(\text{NIR threshold intensity})}{(\text{FUS intensity}) \times (\text{downconversion intensity ratio})} \quad (1)$$

4) Ultrasonic intensity calculation

We can choose the FUS intensity in the target tissue, concerning two conditions.

- Intensity in the range of $<10\text{W/cm}^2$, for larger intensity would harm other tissues.

- One that makes N not too large, for larger N would mean higher shielding effect for UCNPs which would result in poor stimulation.

In this calculation, we choose FUS intensity to be $3W/cm^2$, a appropriate decision fitting both conditions.

5) Calculation of nanoparticle ratios

This choice results in around 100:1 ratio for TD-NP and UCNP near the target neuron. The FUS intensity is $3W/cm^2$, a common choice for US stimulations.

E. Considerations and Improvements

There are some considerations, and room for improvements in this particular method. The first part is nanoparticle size and distribution. TD-NP and UCNP sizes should ensure effective dispersion and minimal immune response. Also liposomes that carry TD-NP/UCNP should be large enough to carry both, but small enough to penetrate the BBB. Also uniform distribution in the targeted brain region is crucial for consistent stimulation. A second point would be timing and delivery. Optimizing the delay between nanoparticle injection and ultrasound application to maximize BBB opening is important to prevent additional infection. Adjust ultrasonic application timing for optimal nanoparticle uptake and light emission is also important. The third part is about nanoparticle ratio and concentration. 100mg/ml is quite a lot amount in the bloodstream, and must be done with caution. This can be improved by using a magnetically-induced liposome or some other way to achieve higher reach rates to the target area, and increasing the lower bounds as given in D.2. The fourth and most important point would be safety and feasibility. We should verify systemic and localized nanoparticle effects to avoid adverse reactions. We should be able to monitor nanoparticle retention and potential long-term effects.

IV. Experimental Design

In order to assess the effectivity of the suggested method, the following factors can be considered.

- Propagation of emitted light in the brain
- Neural stimulation by emitted lights
- Overall delay and efficiency of the *reaction chain*
- Material safety of nanoparticles in human body

Here, the term *reaction chain* refers to the process from the ultrasonic stimulation to the neural stimulation in our method. The former two factors are related to the respective step of the reaction chain. The assessment of those factors, however, have been already covered in previous studies. Considering the novelty of our method resides in the connection between the conversion steps to achieve the optogenetic stimulation, it is sufficient to assess the latter two factors to validate the method.

Parkinson's disease (PD) is a very common neurodegenerative disease, according to [18, Ch. 1]. Although its cause is not clearly identified, the disease is known to present when the dopamine-related neurons are dead in the brain in terms of neuropathology. Optogenetic methods focused on either neurology or therapeutics have been actively studied and discussed as reported in [19]. In following experiments, we validate our non-invasive optogenetic method in mouse models exhibiting PD as showing one example of the effectiveness of the method.

In studies like [20], PD in optogenetics is often dealt by expressing halorhodopsin (HALO) on dopamine neurons through virus vectors. HALO is an anion (chloride) channel protein which is activated by illuminating lights ranging from 525nm to 650nm, representatively described as "yellow light", and it inhibits the neural activity. Once the dopamine neuron is deactivated with the illumination, dopamine release is inhibited, and this process is often used for simulating the symptoms of PD, as it is explained in [21].

Conversely, there are also some research ([22]) on the effectivity of channelrhodopsin-2 (ChR2), a blue-light-activated neuron-activating channel, for improving the PD symptoms. In those studies, ChR2s are expressed on GABA neurons to activate them when illuminated, and it is reported that this optogenetic activation through shows improvement in motor dysfunctions. We focus on this ChR2 activation in the experiments, and many of the in-vivo experimental settings and metrics for the effectivity of the method are also referenced from [22].

A. Preparation

To conduct experiments, it is required to prepare nanoparticle solutions and mouse models with PD. Below are the detailed explanations for the preparation.

1) Nanoparticle Solutions

In the experiments, we basically simulate two different conditions of nanoparticles' presence. This is to test the possibility to minimalize the nanoparticle penetration as long as maintaining the effectivity. Putting it other way, one more option where only UCNPs penetrate the BBB while TD-NPs are flowed through brain blood vessels can also be considered. Here we have in mind the fact that the penetration depth of IR light emitted by TP-NP is longer than that of blue light by UCNP. Therefore, while keeping the same ratio given in the calculation in III.D., 100:1 of TD-NP and UCNP, we prepare the nanoparticles both mixed (condition *a*) and sepataed (condition *b*). The concentration of the nanoparticles also follows the calculation results there. With the solutions. Along with the solutions, ChR2 virus will also be contained.

2) Mouse Models with PD

Adopting the preparation process from [22], we inject 6-hydroxydopamine (6-OHDA), a commonly-used neurotoxin for PD induction in mouse models, into striatum (Str) of brain of the models. A few weeks after the injection, we can determine whether the PD induction is successful by counting tyrosine hydroxylase-immunoreactive (TH^+) neurons in substantia nigra and GABA expressions in zona incerta (ZI), or by measuring cyclomatic numbers, total distance moved, movement velocity, and center zone exploration time of the models. We prepare three groups of the models, for condition *a* and *b* given in 1, and the control (not treated with 6-OHDA).

B. In-vitro Experiments

In-vitro experiments will be conducted to verify our calculations and to estimate the time delay of the reaction chain.

For condition *a*, nanoparticles will be placed in artificial cerebrospinal fluid (aCSF) to simulate in-brain environment. Subsequently, an ultrasonic wave will be applied to the container, and the light emission timing and the spectrum will be analyzed. Meanwhile, for condition *b*, UCNP will still be placed in aCSF, while TD-NP in a fluid simulating blood, for example saline solution mixed with red blood cells. Two different nanoparticle solutions will be separated by a cell culture of endothelial cells representing a blood vessel. US stimulation will then applied to one side where TD-NP solution is present. The analysis will be done in the same manner. In this setting, the appropriate duration of US stimulation for the intended intensity of light emission can be obtained and determined.

C. *In-vivo Experiments*

Based on the estimation obtained in B, we will conduct in-vivo experiments. We can utilize the focused ultrasonic device suggested in [23] for mouse models. A mixture of virus vector and nanoparticle solutions (TD-NP excluded in condition *b*) will be injected into the models, not directly to the brain. Afterwards, through MRI-guided FUS device, the solution will penetrate the BBB. A few wakes later, as in [22], ChR2 channels are expected to be expressed on target GABA neurons. Material safety and biological response is under observation throughout this period. In case of condition *b*, TD-NP is injected later, before FUS stimulation is applied to ZI. The neural activity will be monitored with fast-scan cyclic voltammetry (FSCV), which was used in [13], during the FUS stimulation.

D. *Further Considerations*

As a high portion of nanoparticles that did not penetrate the BBB are expected to circulate in the rest of the body, the effects of nanoparticles should be carefully studied. To reduce the retention, one can consider the liposome targeted drug delivery, as suggested in [24], in the advanced experiments.

Furthermore, if the suggested method here is proven to be effective with PD case, other neurodegenerative disease listed in [19] can be succeeding candidates for the application of this method. Also in context of therapeutics, the devices or methods for lasted stimulation and their effects, e.g. desensitization of ChR2 which was explained in [25], can be discussed in future research.

V. CONCLUSIONS

In this paper, we suggest a novel non-invasive optogenetic method, which incorporates FUS-driven BBB penetration and two nanoparticles. The effectivity of this method would be first validated through in-vivo experiments with mouse models with Parkinson’s disease. The development of a non-invasive optogenetic method holds significant implications not only for research but also for clinical applications. This method provides a safer and more efficient means for functional studies of neurons in specific regions and the treatment of neurological disorders including Parkinson's disease. This research could pave the way for broader and more sustainable applications of optogenetics in research and clinical fields by minimizing the invasiveness of optogenetic techniques.

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